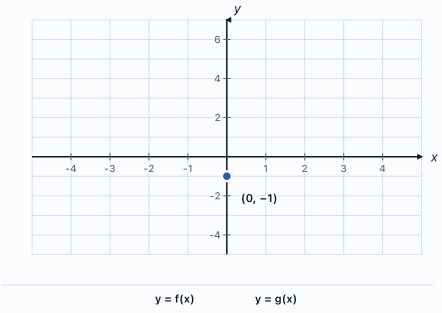
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

GAT Data Interpretation, Averages & Counting Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 12x + 50$ in the domain of GAT Data Interpretation, Averages & Counting. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(8)$.



A 38
B 18
C 30
D 32

2 MCQ

A mathematical model in GAT Data Interpretation, Averages & Counting is defined by the cubic function $y_2(x) = 2x^3 - 6x^2 + 15x - 11$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 33
B 69
C 18
D 21

3 MCQ

In GAT Data Interpretation, Averages & Counting, two displacement vectors are given by $\mathbf{u}_3 = \langle 6, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A −2
B 98
C 100
D 210

4 MCQ

The linear growth profile shown in the diagram for GAT Data Interpretation, Averages & Counting follows an arithmetic progression with first term $a_1 = 7$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 91

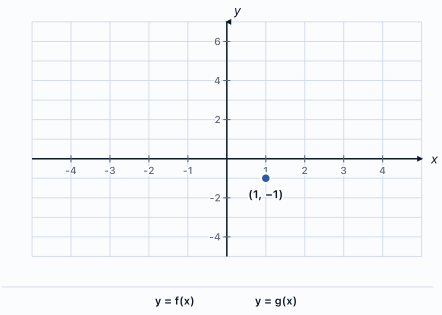
B 825

C 735

D 1365

5 MCQ

In GAT Data Interpretation, Averages & Counting, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 24x)\,\mathrm{d}x$.



A 120

B 120 + 1

C 49

D 96

6 MCQ

In GAT Data Interpretation, Averages & Counting, consider the quadratic function $f_6(x) = x^2 - 14x + 135$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(14)$.

A 221

B 228

C 135

D 214

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 7x^2 + 33x - 18$ studied in GAT Data Interpretation, Averages & Counting. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 417

B 305

C 272

D 67

8 MCQ

In GAT Data Interpretation, Averages & Counting, two displacement vectors are given by $\mathbf{u}_8 = \langle 7, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 21

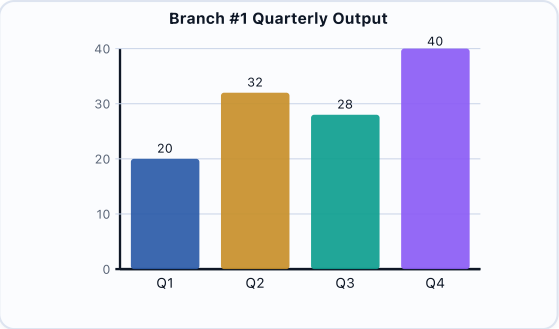
B 98

C 119

D 189

9 MCQ

According to the bar chart shown for Branch #1, the quarterly outputs across Q1, Q2, Q3, and Q4 are 20, 32, 28, and 40 units, respectively. What is the arithmetic mean quarterly output?



A 28

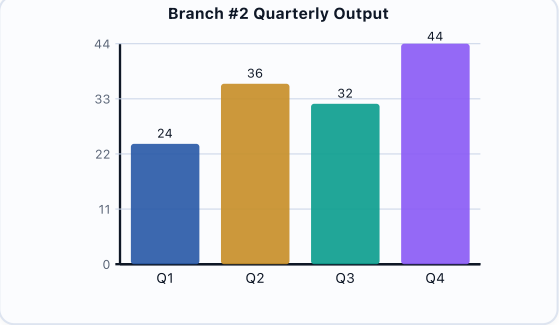
B 30

C 60

D 33

10 MCQ

According to the bar chart shown for Branch #2, the quarterly outputs across Q1, Q2, Q3, and Q4 are 24, 36, 32, and 44 units, respectively. What is the arithmetic mean quarterly output?



A 32

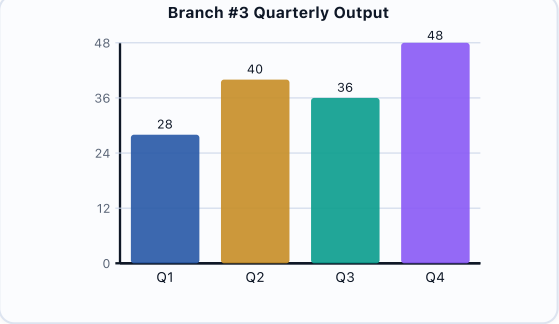
B 34

C 68

D 37

11 MCQ

According to the bar chart shown for Branch #3, the quarterly outputs across Q1, Q2, Q3, and Q4 are 28, 40, 36, and 48 units, respectively. What is the arithmetic mean quarterly output?



A 76

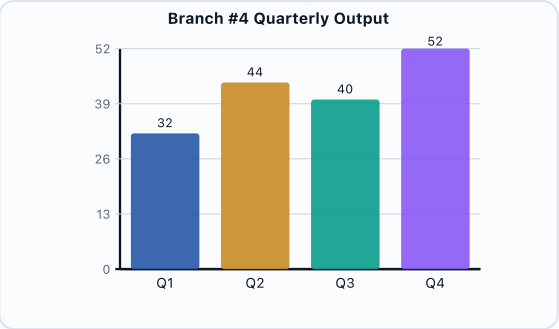
B 41

C 36

D 38

12 MCQ

According to the bar chart shown for Branch #4, the quarterly outputs across Q1, Q2, Q3, and Q4 are 32, 44, 40, and 52 units, respectively. What is the arithmetic mean quarterly output?



A 84

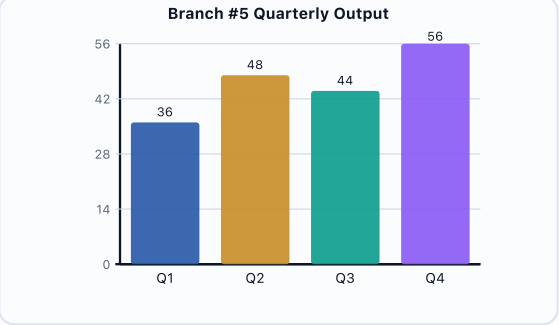
B 40

C 42

D 45

13 MCQ

According to the bar chart shown for Branch #5, the quarterly outputs across Q1, Q2, Q3, and Q4 are 36, 48, 44, and 56 units, respectively. What is the arithmetic mean quarterly output?



A 44

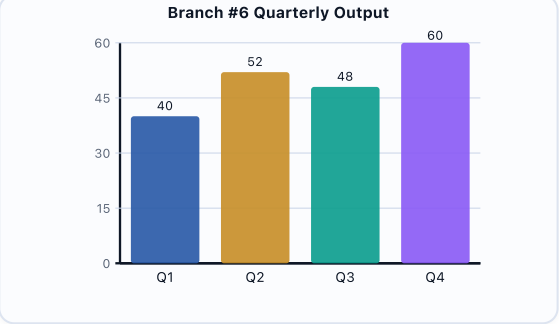
B 46

C 92

D 49

14 MCQ

According to the bar chart shown for Branch #6, the quarterly outputs across Q1, Q2, Q3, and Q4 are 40, 52, 48, and 60 units, respectively. What is the arithmetic mean quarterly output?



A 53

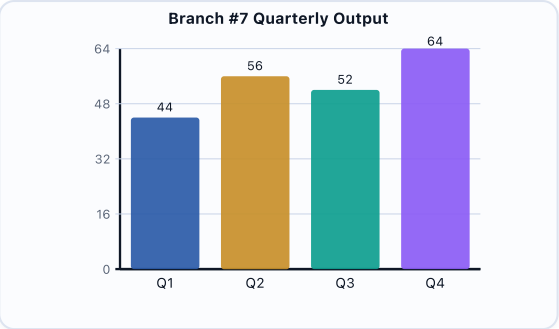
B 50

C 48

D 100

15 MCQ

According to the bar chart shown for Branch #7, the quarterly outputs across Q1, Q2, Q3, and Q4 are 44, 56, 52, and 64 units, respectively. What is the arithmetic mean quarterly output?



A 57

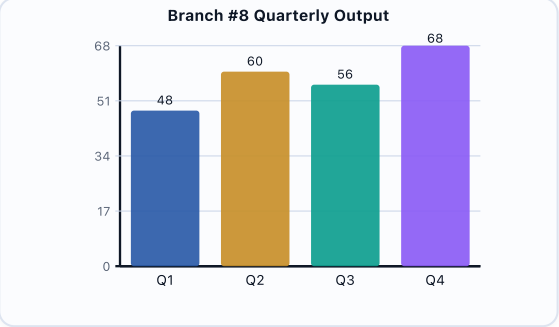
B 54

C 52

D 108

16 MCQ

According to the bar chart shown for Branch #8, the quarterly outputs across Q1, Q2, Q3, and Q4 are 48, 60, 56, and 68 units, respectively. What is the arithmetic mean quarterly output?



A 116

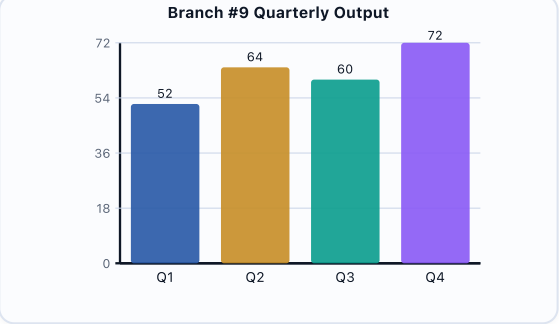
B 58

C 56

D 61

17 MCQ

According to the bar chart shown for Branch #9, the quarterly outputs across Q1, Q2, Q3, and Q4 are 52, 64, 60, and 72 units, respectively. What is the arithmetic mean quarterly output?



A 62

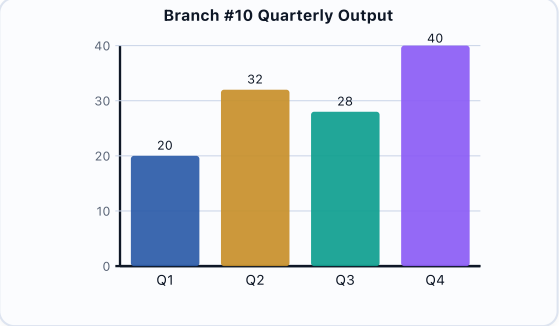
B 65

C 60

D 124

18 MCQ

According to the bar chart shown for Branch #10, the quarterly outputs across Q1, Q2, Q3, and Q4 are 20, 32, 28, and 40 units, respectively. What is the arithmetic mean quarterly output?



A 30

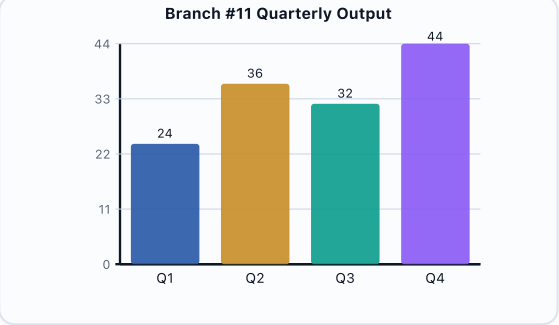
B 60

C 33

D 28

19 MCQ

According to the bar chart shown for Branch #11, the quarterly outputs across Q1, Q2, Q3, and Q4 are 24, 36, 32, and 44 units, respectively. What is the arithmetic mean quarterly output?



A 68

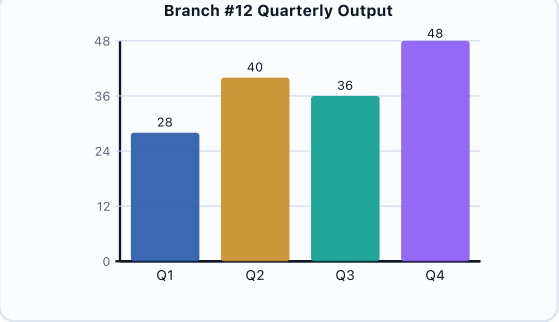
B 32

C 37

D 34

20 MCQ

According to the bar chart shown for Branch #12, the quarterly outputs across Q1, Q2, Q3, and Q4 are 28, 40, 36, and 48 units, respectively. What is the arithmetic mean quarterly output?



A 76

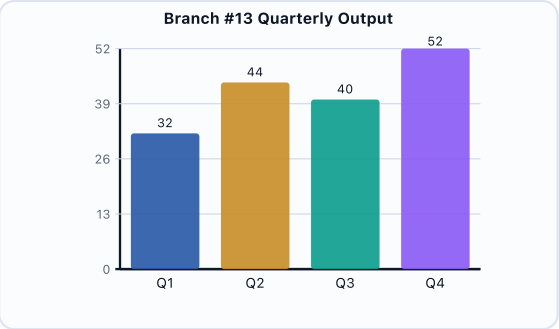
B 41

C 36

D 38

21 MCQ

According to the bar chart shown for Branch #13, the quarterly outputs across Q1, Q2, Q3, and Q4 are 32, 44, 40, and 52 units, respectively. What is the arithmetic mean quarterly output?



A 40

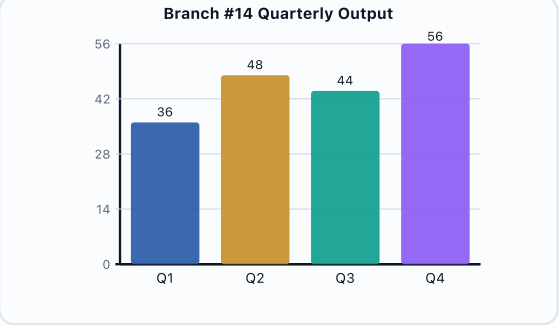
B 84

C 42

D 45

22 MCQ

According to the bar chart shown for Branch #14, the quarterly outputs across Q1, Q2, Q3, and Q4 are 36, 48, 44, and 56 units, respectively. What is the arithmetic mean quarterly output?



A 49

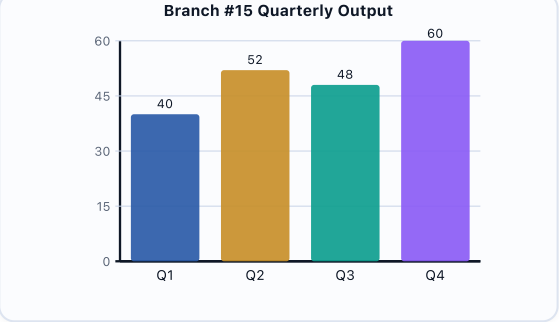
B 92

C 44

D 46

23 MCQ

According to the bar chart shown for Branch #15, the quarterly outputs across Q1, Q2, Q3, and Q4 are 40, 52, 48, and 60 units, respectively. What is the arithmetic mean quarterly output?



A 48

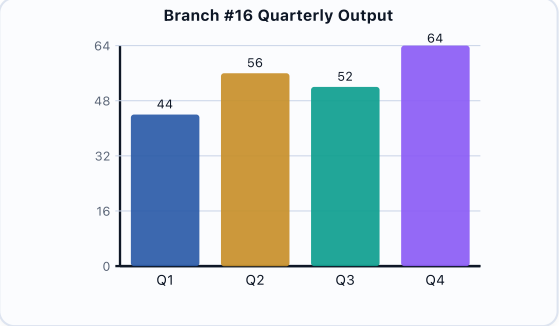
B 53

C 50

D 100

24 MCQ

According to the bar chart shown for Branch #16, the quarterly outputs across Q1, Q2, Q3, and Q4 are 44, 56, 52, and 64 units, respectively. What is the arithmetic mean quarterly output?



A 54

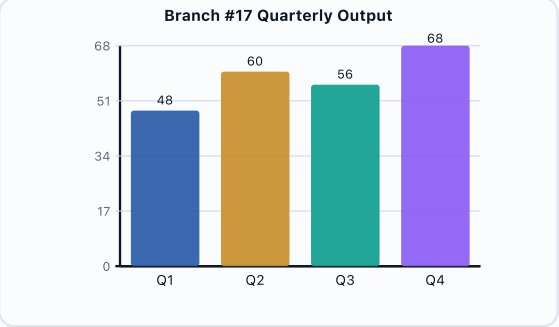
B 57

C 52

D 108

25 MCQ

According to the bar chart shown for Branch #17, the quarterly outputs across Q1, Q2, Q3, and Q4 are 48, 60, 56, and 68 units, respectively. What is the arithmetic mean quarterly output?



A 56

B 61

C 116

D 58